INCH-POUND

A-A-55538B
26 September 2023
SUPERSEDING
A-A-55538A
19 AUGUST 2013

COMMERCIAL ITEM DESCRIPTION

WATERPROOF SPARK PLUG LEAD AND CONDUIT ASSEMBLY (OTHER THAN AIRCRAFT)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for waterproof spark plug lead and conduit assembly (other than aircraft). Waterproof spark plug lead and conduit assembly (other than aircraft) covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION/PART OR IDENTIFIYING NUMBER (PIN). This CID uses a classification system which is included in the PIN as shown in the following example (see 7.1).



CID number

Dash number
(see 3.2 & Table I)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: DLA Land and Maritime, ATTN: VAI, P.O. Box 3990, Columbus OH 43218–3990, or email fluidflow@.dla.mil. Since contact information can change you may want to verify the currency of the address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/C

FSC 2920

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3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Waterproof spark plug lead and conduit assembly (other than aircraft) supplied to this CID shall be as specified herein (see Figure 1).

3.2 Assemblies. This CID covers 2 types of assemblies. This CID uses a classification system which includes the PIN as shown in Table I.

3.3 Length. The nominal length A shown on Figure 1, shall be as specified in Table I.

TABLE I. Dash numbers. 1/

AA55538 CID dash number	Type	CID dash numbers length A, (See Figure 1)	
No.		Inches	Millimeters
002	I	10 ± 1/8	254 ± 3
004		12 ± 1/8	305 ± 3
006		17 ± 1/4	432 ± 6
008		20 ± 1/4	508 ± 6
011		26 ± 1/4	660 ± 6
012		29 ± 1/2	737 ± 13
001	II	8 ± 1/8	203 ± 3
003		10 ± 1/8	254 ± 3
005		14 ± 1/4	356 ± 6
007		17 ± 1/4	432 ± 6
009		20 ± 1/4	508 ± 6
010		23 ± 1/4	584 ± 6
013		32 ± 1/2	813 ± 13
014		40 ± 1/2	1016 ± 13
015		45 ± 1/2	1143 ± 13
016		50 ± 1/2	1270 ± 13
017		55 ± 1/2	1397 ± 13

1/ NOTES:

1. Type I Connector is straight (inline)
2. Type II - Connector has a 90° elbow soldered to the conduit.
3. Length A - Nominal length is defined as the distance between connecting nuts (for Type I connectors), or the distance between the inside of the soldered joint at the elbow and the connecting nut at the opposite end (for Type II connectors). See Figure I for details.
4. CID dash numbers came from basic CID to provide cross reference.

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3.4 Connector configuration. The lead and conduit assemblies described in this CID include both straight (inline) connectors or connectors that contain a 90° elbow as specified in Table I and shown on Figure 1.

3.5 Marking. Waterproof spark plug lead and conduit assembly (other than aircraft) supplied to this CID shall be marked with the manufacturer's (MFR's) standard commercial PIN. (NOTE: The part number marked on the unit pack shall be the CID PIN.)

3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.7 Workmanship. Waterproof spark plug lead and conduit assembly (other than aircraft) shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. REGULATORY REQUIREMENTS. The offeror/Contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company producing the item must have sold 1,000 units meeting this CID in the commercial marketplace over the past 2 years.

5.3 Certification. Certification must be done with the procuring activity approval. The Contractor shall certify that the product offered meets the salient characteristics of the description and conforms to the producer’s own drawings, specifications, standards, and quality assurance practices, and is the same as the product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

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7.2 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website https://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see Section 3).

7.3 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these waterproof spark plug lead and conduit assembly (other than aircraft) to DLA Land and Maritime under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.4 Source of documents.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52440	–	Conduit, Metal, Flexible: Electrical Shielded
A-A-59588	–	Rubber, Silicone

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-21425	–	Wire, Steel, High Carbon, Square, Uncoated, For Mechanical Springs (General Purpose)

(Copies of these documents are available online at https://quicksearch.dla.mil or from the DLA Document Services Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094)

FEDERAL REGULATIONS

FAR	–	Federal Acquisition Regulations (FAR)

(Copies of these documents are available online at www.acquisition.gov/comp/far/index.html or from the U.S. Government Printing Office, 732 North Capital Street, NW, Washington D.C. 20401.)

OTHER PUBLICATIONS

ASTM INTERNATIONAL

ASTM A313/A313M	–	Standard Specification for Stainless Steel Spring Wire
ASTM A575	–	Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades
ASTM A580/A580M	–	Standard Specification for Stainless Steel Wire
ASTM A663/A663M	–	Standard Specification for Steel Bars, Carbon, Merchant Quality, Mechanical Properties
ASTM B16/B16M	–	Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines
ASTM B121/B121M	–	Standard Specification for Leaded Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B124/B124M	–	Standard Specification for Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B36/B36M	–	Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar

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ASTM B700 – Standard Specification for Electrodeposited Coatings of Silver for Engineering Use

(Copies of these documents are available online at https://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE-AMS-QQ-P-416 – Plating, Cadmium (Electrodeposited)

(Copies of these documents are available on line at https://www.sae.org from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

7.5 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.

b. Product conformance provisions.

c. Packaging requirements.

7.6 Commercial products. As part of the market analysis and research effort, this CID was coordinated with the following manufacturers of commercial products. At the time of CID preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict acquisition to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
00642	Heale Manufacturing co inc 1231 The Strand Waukesha, WI 53186-3861 http://www.healemfg.com (262) 542-4496 Elliott Erickson John Livingston govt@healemfg.com
4F040	Pitt Auto Electris Company 4085 Alpha Dr. Allison Park, Pa 15101-2961 (412) 487-5075 Bill Anker bill@pittauto.com Dick Eiseman dick@pittauto.com

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7.7 Government users. To acquire information on obtaining these waterproof spark plug lead and conduit assembly (other than aircraft) from the Government inventory system, contact DLA Land and Maritime, ATTN DLA Land and Maritime Call Center (–NAB), P.O. Box 3990, Columbus, OH 43218–3990 or telephone (614) 692-2271 or (614) 692-3191.").

7.8 Legacy. This commercial item description is a replacement for MS51010 and MS51011 for all federal agencies (MS51010 is canceled as of 10 October 1990 and MS51011 is canceled as of 3 November 1966 copies of these documents are available online at https://quicksearch.dla.mil

7.9 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and Contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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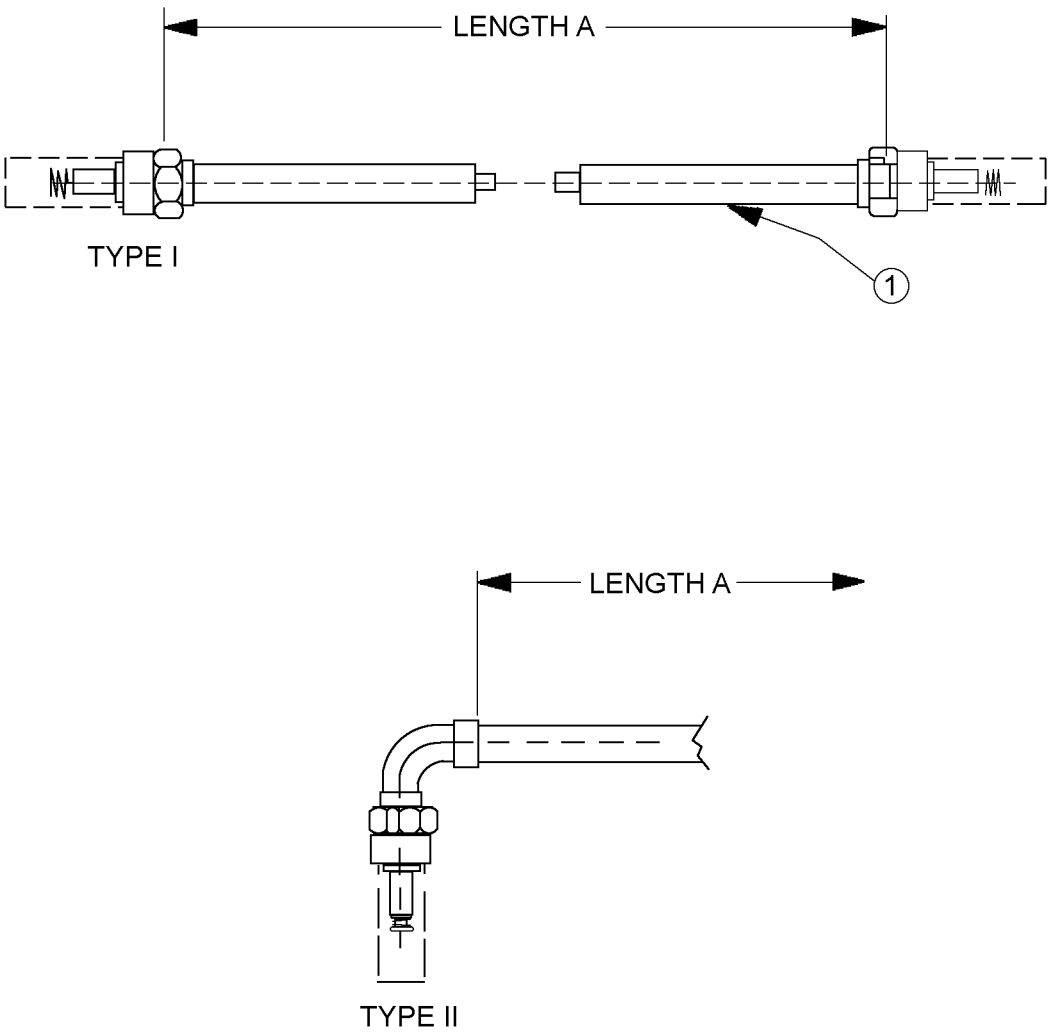
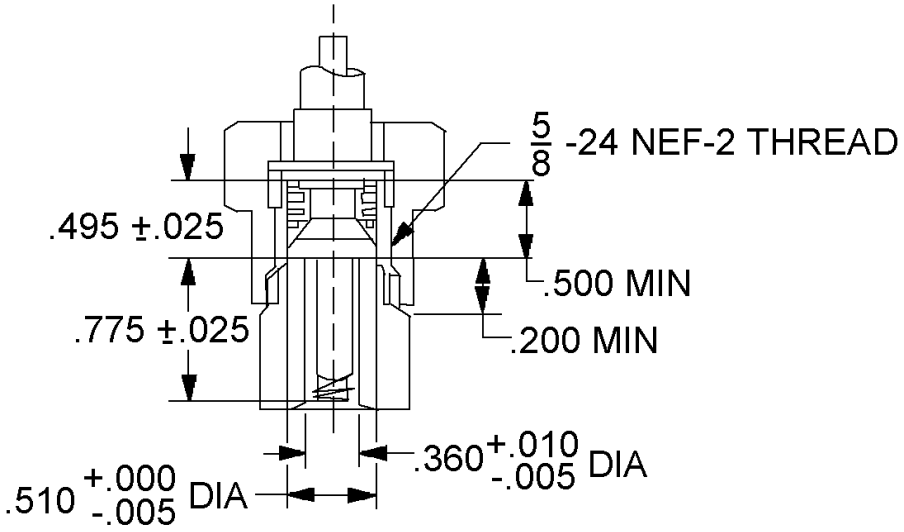
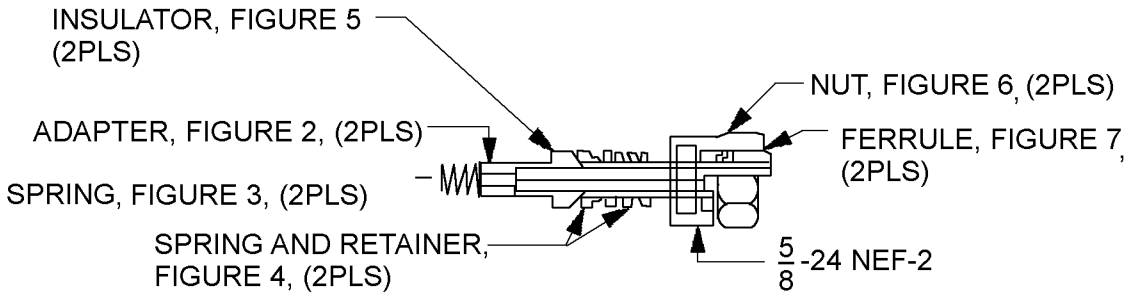


FIGURE 1. Waterproof sparks plug lead and conduit assembly.

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VIEW A
TERMINAL END INSTALLATION

FIGURE 1. Waterproof spark plug lead and conduit assembly -continuous.

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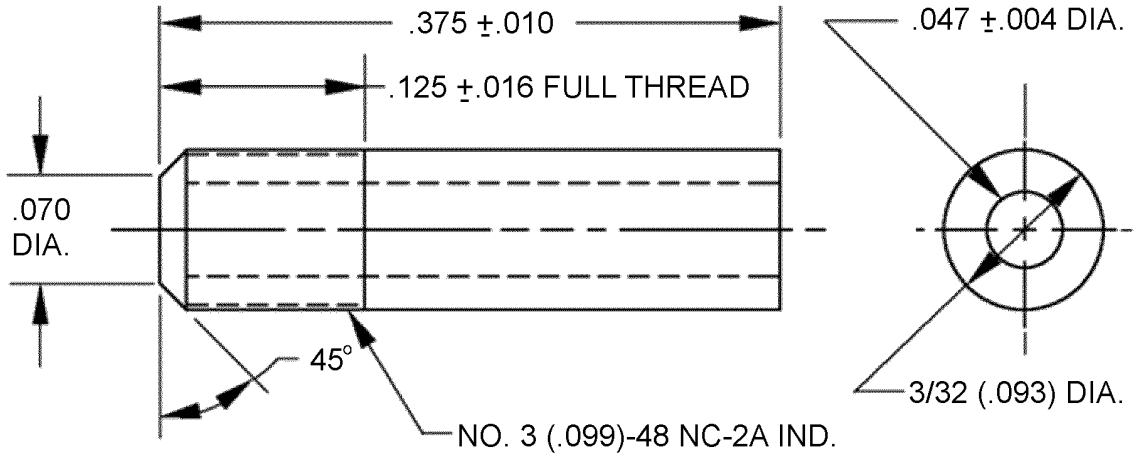
Inch	mm	Inch	mm
.005	.13	.430	10.92
.010	.25	.495	12.57
.015	.381	.500	12.70
.025	.64	.510	12.95
.200	5.08	.775	19.69
.281	7.14		
.306	7.78		

NOTES:

1. Conduit: OD., .430 ±.010. ID., .281±.015, A-A-52440, type I, grade B, with single layer, two-over-two weave, braid.
2. Metrics are information only.

FIGURE 1. Waterproof sparks plug lead and conduit assembly – continuous.

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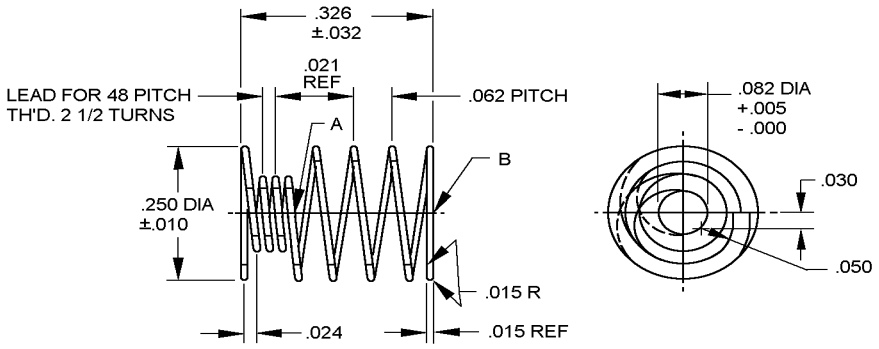
Inch	mm
.004	.102
.010	.254
.016	.406
.070	1.78
.047	1.19
.093	2.36
.125	3.18
.375	9.53

NOTES:

1. Material: Brass as specified in ASTM B16/B16M, ASTM B36/B36M, ASTM B121/B121M and ASTM B124/B124M.
2. Finish: Silver as specified in ASTM B700.
3. Tolerance: +/-.005, +/- 1/64, +/- , 2°. Unless otherwise specified.
4. Dimensions are in inches. Metric equivalents are for information only.

FIGURE 2. Adapter Lead, to terminal: Spark Plug cable (other than aircraft).

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Inch	mm	Inch	mm
.003	.076	.032	.813
.005	.127	.050	1.27
.010	.254	.062	1.57
.015	.381	.082	2.08
.021	.533	.250	6.35
.024	.610	.326	8.28
.030	.762		

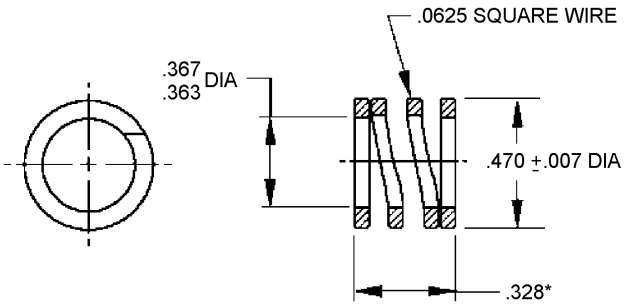
DIMENSIONS ARE IN INCHES

NOTES:

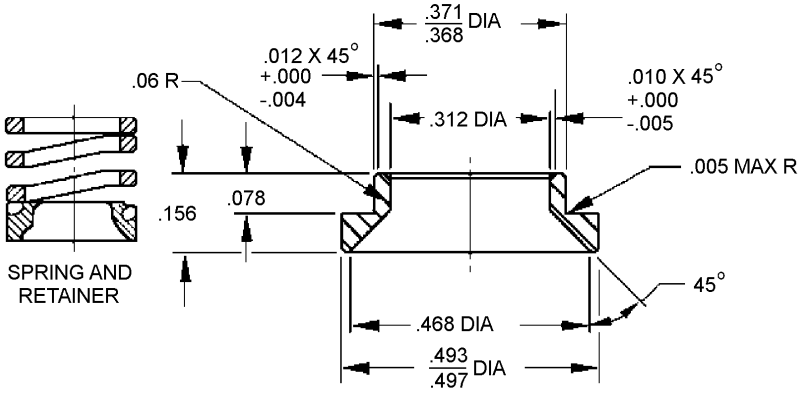
1. Material: Steel as specified in: ASTM A313/A313M and ASTM A580/A580M, spring temper.
2. Finish: Passivation.
3. Tolerance: +/-.005, +/- 1/64, +/- , 2º, Unless otherwise specified.
4. Wire cross section shall be: .015 +/- .003.
5. Spring turn shall be: 8.5 turns, both ends shall be ¾ turn squared.
6. Lengths: Between points A and B is compressible solid under a 1 pound load.
7. Dimensions are in inches. Metric equivalents are for information only.

FIGURE 3. Spring, helical, compression spark plug cable terminal (other than aircraft).

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DETAIL OF SPRING



DETAIL OF RETAINER

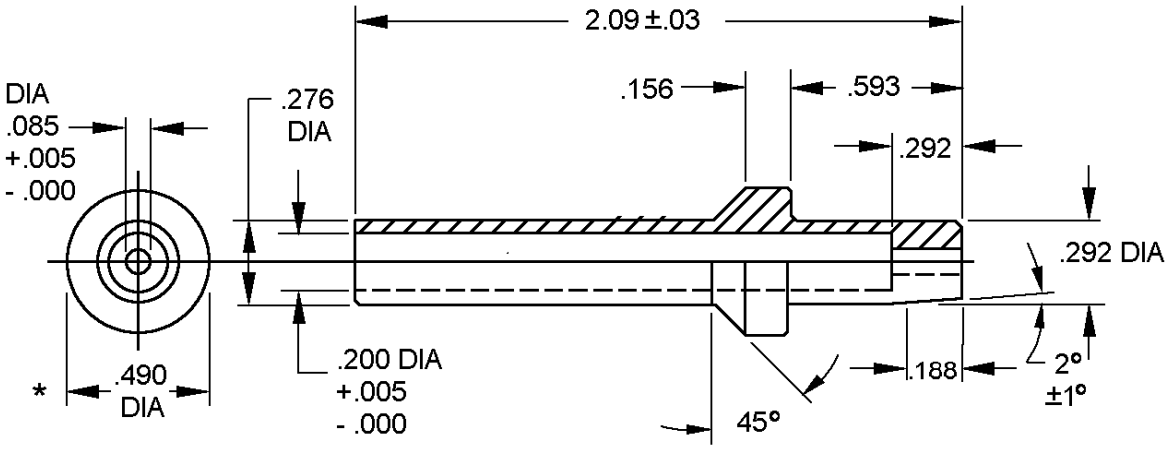
Inch	mm	Inch	mm	Inch	mm
.004	.102	.0625	1.59	.367	9.32
.005	.127	.070	1.78	.368	9.35
.006	.152	.078	1.98	.469	11.89
.010	.254	.156	3.96	.493	12.52
.012	.305	.312	7.92	.495	12.57
.030	.762	.328	8.33	12.57	319.3
.060	1.52	.363	9.22		

NOTES:

1. Material: Spring steel wire .0625 square, as specified in MIL-W-21425, type I.
2. Material Retainer: Carbon Steel: ASTM A575 and ASTM A663/A663M.
3. Finish: Cadmium plating as specified in SAE-AMS-QQ-P-416, class 3.
4. within 1 hour after plating spring and retainer be baked for 2 hours at 375° F +/-15°.
5. Spring ends: Closed and ground square: 2-1/2 effective coils, compressible to 1/4 inch overall length. under 15 +/-3 pound load.
6. Tolerance: +/-.005, +/- 1/64, +/- , 2°. Unless otherwise specified.
7. Dimensions are in inches. Metric equivalents are for information only

FIGURE4. Spring and retainer, electrical terminal: spark plug cable (other than aircraft).

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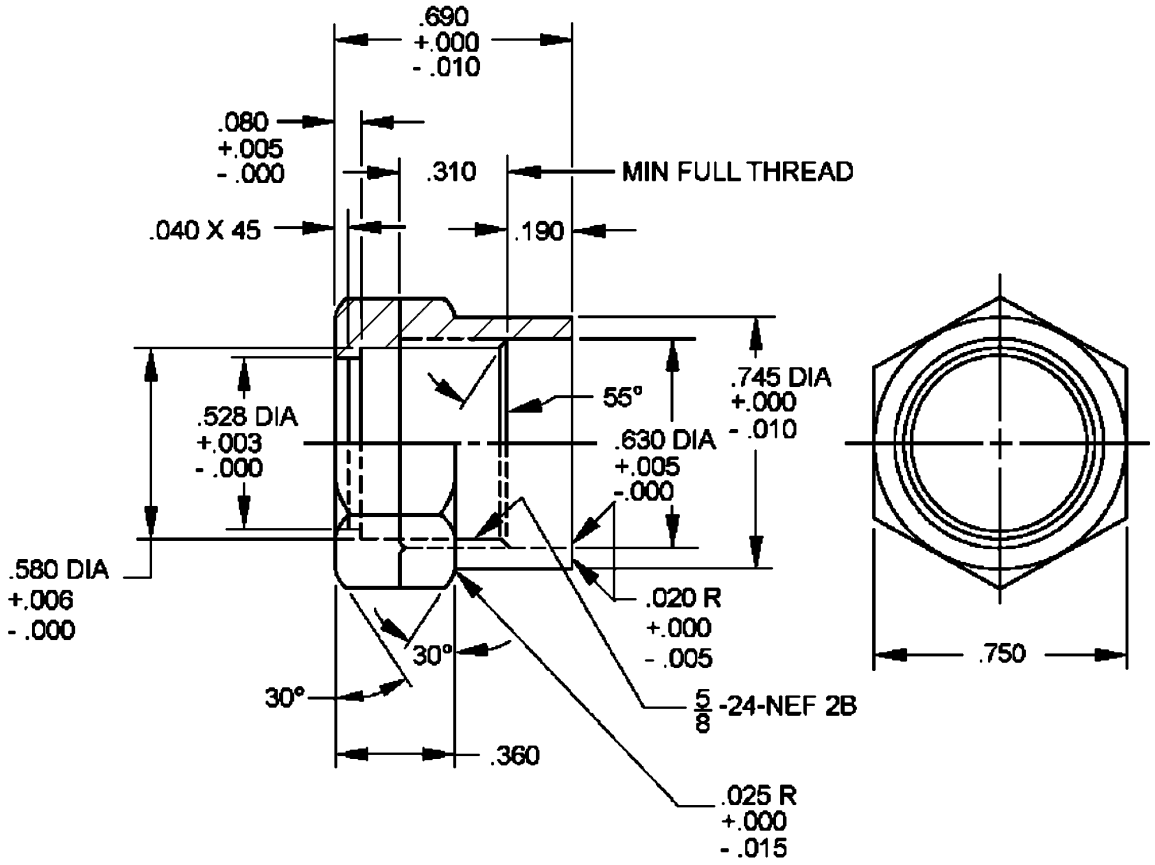
Inch	mm	Inch	mm
.005	.127	.276	7.01
.030	.762	.292	7.42
.085	2.16	.490	12.45
.156	3.96	.593	15.06
.188	4.78	2.09	53.09
.200	5.08		

NOTES:

1. Material. Silicone rubber as specified in A-A-59588, class 2B, grade 80.
2. Finish. Part must free of mold release compound.
3. Fillet radii. All fillet radii .030 maximum. .Unless otherwise specified.
6. Tolerance: +/-.005, +/- 1/64, +/- 2°. Unless otherwise specified.
7. Dimensions are in inches. Metric equivalents are for information only

FIGURE 5. Insulator, bushing: spark plug cable terminal (other than aircraft).

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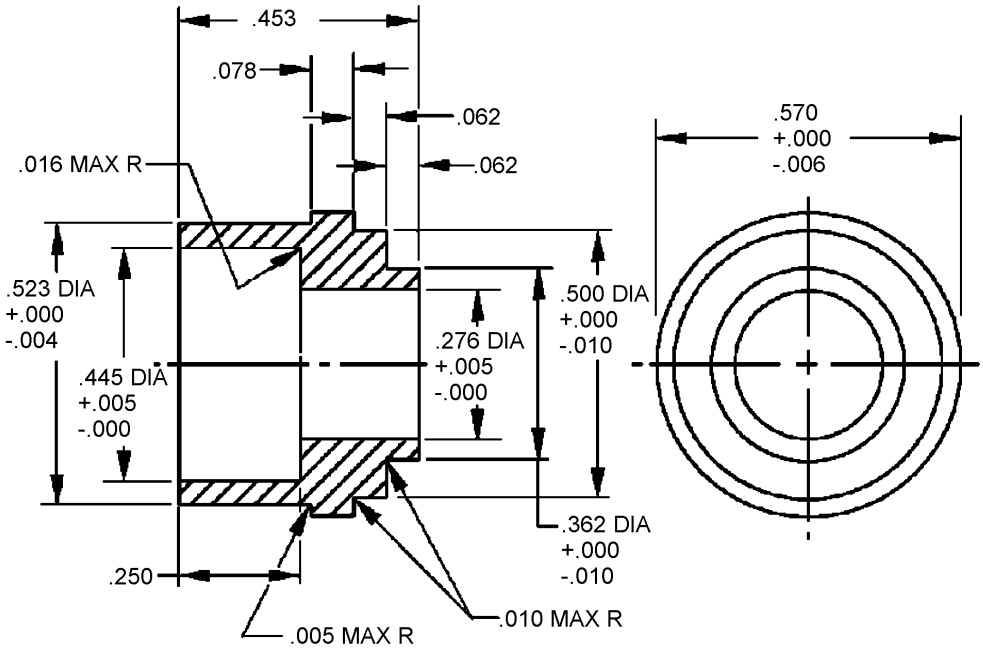
Inch	mm	Inch	mm
.003	.076	.360	9.14
.005	.127	.528	13.41
.006	.152	.580	14.73
.010	.254	.630	16.00
.015	.381	.690	17.53
.020	.508	.745	18.92
.025	.635	.750	19.05
.040	1.02		
.080	2.03		
.190	4.83		
.310	7.87		

NOTES:

1. Material. Steel, carbon, C-1015, to C1025 as specified in ASTM A575 and ASTM A663/A663M.
2 Finish. Cadmium plating as specified in SAE-AMS-QQ-P-416, type II, class2.
3. Tolerance. +/-.005, +/- 1/64, +/- 2°. Unless otherwise specified.
4. Dimensions are in inches. Metric equivalents are for information only.

FIGURE 6. Nut, coupling, electrical conduit sparks plug cable (other than aircraft).

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Inch	mm	Inch	mm
.005	.127	.276	7.01
.006	.152	.362	9.19
.010	.254	.445	11.30
.016	.406	.453	11.51
.062	1.57	.500	12.7
.078	2.00	.523	13.28
.250	6.35	.570	14.48

NOTES:

1. Material: Brass as specified in ASTM B16/B16M, ASTM B36/B36M, ASTM B121/ B121M,
and ASTM B124/B124M.
2 Finish. Cadmium plating as specified in SAE-AMS-QQ-P-416, type II, class2.
3. Tolerance. +/-.005, +/- 1/64, +/- 2°. Unless otherwise specified.
4. Dimensions are in inches. Metric equivalents are for information only

FIGURE 7. Ferrule, electrical conduit spark plugs cable (other than aircraft).

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MILITARY INTERESTS

Custodians:
Army - AT
Navy - YD
Air Force - 71
DLA - CC

Review activities:
Army-CR, GL
Navy – MC, SA

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FAS

Preparing activity:
DLA - CC

Project 2920-2023-005

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